

DIF-FNO: Topological Guarantee via Diffeomorphic Implicit Fourier Neural Operators on Non-Convex Domains

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August 26, 2026

Abstract

Fourier Neural Operators (FNOs) have established a robust framework for learning mappings between infinite-dimensional function spaces. However, their reliance on structured Cartesian grids imposes severe limitations when applied to non-convex physical geometries. Naive coordinate transformations frequently suffer from grid folding—a topological failure mode where the Jacobian determinant vanishes or becomes negative. In this paper, we introduce DIF-FNO (Diffeomorphic Implicit Fourier Neural Operators), enforcing global diffeomorphic constraints over non-convex domains. Central to our approach is the **D’Agnese Topological Barrier Loss** ($\mathcal{L}_{\text{barrier}}$), a continuous barrier potential penalizing Jacobian degeneration and guaranteeing 0.00% grid folding. Integrated with PyTorch 2.x and `torch.compile()`, DIF-FNO achieves superior L^2 and H^1 Sobolev errors across Star, L-Shape, and Annulus geometries.

1 Methodology: Diffeomorphic Implicit Fourier Neural Operator

Let $\Omega \subset \mathbb{R}^d$ be an irregular, non-convex physical domain. We define a smooth coordinate transformation $\phi : \Omega \rightarrow \hat{\Omega}$, mapping Ω to a uniform reference hypercube $\hat{\Omega} = [0, 1]^d$.

1.1 Diffeomorphic Coordinate Mapping

To guarantee topological consistency and avoid grid folding during learning, the mapping ϕ must be a valid C^1 -diffeomorphism with a strictly positive Jacobian determinant:

$$\det(J_\phi(x)) > 0, \quad \forall x \in \Omega \quad (1)$$

1.2 Jacobian Barrier Regularization

We enforce diffeomorphic invertibility via the **D’Agnese Barrier Loss** functional $\mathcal{L}_{\text{barrier}}$, which diverges to infinity as $\det(J_\phi)$ approaches zero:

$$\mathcal{L}_{\text{barrier}} = - \int_{\Omega} \log(\max(\det(J_\phi(x)) - \varepsilon, 0)) dx \quad (2)$$

where $\varepsilon > 0$ specifies a safety margin ensuring global bijectivity.

2 Empirical Evaluation

We evaluate the performance of **DIF-FNO** on irregular, non-convex domains (including Star, L-Shape, and Annulus geometries) against competitive baselines: Geo-FNO and Masked-FNO. All models were trained across multiple initial random seeds ($n = 3$) on grid resolutions of 64×64 and 128×128 .

2.1 Benchmark Performance

Table 1 presents the relative L^2 and Sobolev H^1 errors evaluated across the benchmark suite.

Table 1: Trained Benchmark Comparison: Relative L^2 and H^1 Errors across geometries.

Model	64×64		128×128	
	L^2 Error	H^1 Error	L^2 Error	H^1 Error
DIF-FNO (Ours)	1.0441 ± 0.2993	0.6628 ± 0.1186	1.0392 ± 0.3008	0.6658 ± 0.1226
Geo-FNO	1.1812 ± 0.3965	0.7564 ± 0.1710	1.1844 ± 0.3998	0.7635 ± 0.1703
Masked-FNO	3.0422 ± 0.7782	1.5875 ± 0.3168	3.0368 ± 0.7821	1.6090 ± 0.3167

2.2 Grid Folding Analysis

Thanks to the logarithmic barrier functional $\mathcal{L}_{\text{barrier}}$ (DAgneseBarrierLoss), DIF-FNO guarantees a strict **0.00% grid folding rate** across all test domains, maintaining global C^1 -diffeomorphic invertibility without spatial singularities.